A sharp threshold for three-colour targets from four-colour cubes

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Abstract

Any collection of 29 cubes, each displaying all four colours of a fixed palette, contains eight that form a $2 \times 2 \times 2$ cube with monochromatic faces using exactly three colours. There is a collection of 28 cubes with no such solution. This answers the threshold question in Problem 2 of Berkove, Eickemeyer and Schulman (2018). Moreover, there are exactly two labelled 28-cube obstructions, exchanged by reflection. The proof uses the corner-containment reduction of Berkove, Eickemeyer and Schulman and elementary counting. The threshold and classification are formalised in Lean.

AI transparency statement. The research argument, computer code, Lean formalisation and manuscript were developed using AI under the author's direction. The declaration at the end gives the scope of this work and its review status.

1 Introduction

Berkove, Eickemeyer and Schulman [1, Problem 2] ask for the least number of four-colour cubes that guarantees a subset of eight forming a three-colour corner solution: a $2 \times 2 \times 2$ cube whose exterior faces are monochromatic and use exactly three colours. Here every source cube uses all four colours of a fixed palette, and repeated varieties are allowed.

We prove that the least number is 29 and classify all collections of 28 cubes that fail. There are exactly two such collections when colour labels are fixed, and reflection exchanges them. Each consists of seven copies of each of four varieties; explicit face tuples are given in Theorem 3. The proof first classifies the possible 28-cube obstructions by corner containment and counting, then deduces the upper bound and verifies that both obstructions fail.

2 Model and geometric facts

Colours are fixed, varieties are identified by proper rotations, and reflections are not allowed as moves. Collections are multisets: repeated varieties are allowed. Thus "labelled" refers to the colour labels, not to labels on individual copies of a cube. Only exterior faces must match the target; internal faces are unrestricted.

A corner is a cyclically ordered triple. Read its face colours counterclockwise as viewed from outside along the corner's body diagonal; equivalently, order the three outward face normals to form a positively oriented basis. Thus abc and acb are distinct when the three colours differ. A source cube fits a target corner by a proper rotation if and only if it has the same cyclic corner type: the cyclic alignments of two corner frames are exactly those realised by proper rotations of the cube. A cube supplies one corner position, regardless of how many suitable corners it has. For matching purposes, a target is represented by its multiset of eight corner types: a collection builds it exactly when these eight occurrences can be matched to distinct cubes having the required types.

Write $Q(a;b,c)$ for a target whose b and c faces are opposite and whose other four faces have colour a. Its corner multiset is $4aab+4aac$. Write $R(a;b,c)$ for the analogous target with b and c adjacent; its corners are

$$2aaa+2aab+2aac+abc+acb.$$

Finally, a target with opposite pairs aa, bb, cc has four corners of each orientation on $\{a,b,c\}$. We need only these three families of exactly-three-colour targets for the sufficient constructions.

If every corner type of a variety v occurs on a variety w, replacing a copy of w by v preserves unsolvability: each small cube is used at just one corner. This is the reduction of [1, Section 2]. For clarity, Table 1 gives the entire reduction needed here. A face tuple is read in the order $+x,-x,+y,-y,+z,-z$. The rows represent the seven orbits under colour permutations, with rotations already identified. The three non-retained representatives all contain the corner types of 001123. Colour permutations transport these containments to every member of their orbits.

Representative	Orbit size	Retained type	Replacement
001123	6	checkerboard C	—
010203	8	triple-corner T	—
010213	12	pair type P	—
010223	12	pair type P	—
000123	12	—	001123
001213	12	—	001123
010123	6	—	001123

Table 1: The 68 varieties reduce to 38, with colour labels fixed.

The retained types are precisely the minimal varieties in [1, Table 8]. A reduced variety belongs to bucket a if it has at least two of the three corner types aab with $b \neq a$. We use the following explicit description.

- The eight T varieties have a monochromatic corner aaa, and also aab for each of the other three colours b. Each belongs to the bucket of its repeated colour a.
- The 24 P varieties belong to two buckets, one for each repeated colour. In bucket a, such a cube has exactly two of the three types aab with $b \neq a$.
- The six checkerboard varieties have two repeated colours on opposite pairs of faces and two singleton colours on the remaining opposite pair. A checkerboard whose repeated colours are a, b has both orientations on the triples $\{a,b,c\}$ and $\{a,b,d\}$. It belongs to no bucket.

The 32 ordinary varieties (T and P) split into two classes of 16. Every member of one class has the same four trichromatic corner types, one orientation for each three-element subset of the palette; every member of the other class has the four reversed orientations. Each class contains exactly one T variety for each repeated colour. For our coordinate convention the four types in one class are

$$\Omega = \{012, 031, 023, 132\}.$$

Table 2 lists the corner types of the four retained representatives. These sets and their colour permutations give the stated incidences and orientation classes. A separate geometric checker verifies both tables and all the stated incidences using physical rotations.

Representative	Corner-type set
001123	$\{012, 013, 021, 031\}$
010203	$\{000, 001, 002, 003, 012, 023, 031, 132\}$
010213	$\{001, 003, 011, 012, 023, 031, 112, 132\}$
010223	$\{002, 003, 012, 022, 023, 031, 122, 132\}$

Table 2: Corner types of the retained representatives, with cyclic triples written in lexicographically least form. Multiplicities are omitted: each source cube supplies only one target position.

Lemma 1 (Bucket bound)**.** *Eight reduced cubes in a single bucket can build a three-colour target. Consequently, a reduced collection with no such solution has at most seven cubes in each bucket.*

Proof. Each of the eight cubes misses at most one of the three types *aab*. Thus at most one type is missed by five or more cubes. Choose two types, *aab* and *aac*, each missed by at most four cubes. Assign cubes missing *aac* to *aab* positions, and cubes missing *aab* to *aac* positions. These groups are disjoint and each has size at most four. Every remaining cube supports both types, so they fill the remaining positions of $Q(a; b, c)$. □

3 The extremal collections

Theorem 1. *A collection of 28 four-colour cubes without a three-colour corner solution consists of seven copies of each of the four T varieties in one ordinary orientation class.*

Proof. Reduce the collection as above and suppose that no solution exists. Let $A \geq B$ be the sizes of the two ordinary orientation classes, and let C count checkerboards, so $A + B + C = 28$. Call the class of size A the majority class. Call the remaining $M = B + C$ cubes *minority cubes*.

For a three-element colour set E, let z_E count checkerboards supporting its two orientations. If $z_E \geq 8$, those eight checkerboards build the target with three monochromatic opposite pairs. Thus $z_E \leq 7$ for each E, and, since each checkerboard supports two triples, $2C \leq 28$. It follows that $A + B \geq 14$, so $A \geq 7$. There are $m_E = B + z_E$ minority cubes supporting the orientation on E opposite to that of the majority class. If $m_E \geq 4$, four such cubes and four majority cubes build the target with three monochromatic opposite pairs. Hence $m_E \leq 3$ for every E. Summing gives

$$4B + 2C \leq 12, \qquad M \leq 6, \qquad A = 28 - M \geq 22. \tag{1}$$

From now on, let t_a and p_a count only majority-class T and P cubes in bucket a, and let n_T, n_P be their respective totals, so $A = n_T + n_P$. By Lemma 1, $t_a + p_a \leq 7$. Each T cube belongs to one bucket and each P cube to two, so summing gives

$$n_T + 2n_P \leq 28, \qquad n_P \leq M. \tag{2}$$

We will repeatedly use the following construction. If six majority cubes supply $2aaa + 2aab + 2aac$ and a minority cube supports $\{a, b, c\}$, then the minority cube supplies the opposite trichromatic orientation and an unused majority cube supplies the other. Such a cube is available since $A \geq 22$. This builds $R(a; b, c)$.

Suppose $M > 0$. Every minority cube supports some triple containing any given colour a. The construction above therefore rules out $t_a \geq 6$. Thus $t_a \leq 5$ for every a, and $n_T \leq 20$. Since $n_T + n_P + M = 28$ and $n_P \leq M$, we obtain $M \geq 4$.

For any two distinct colours a, b, some minority cube supports a triple containing both. Otherwise every minority cube would have to be a checkerboard with the complementary pair of repeated colours. One of the corresponding m_E would then equal $M \geq 4$, contradicting $m_E \leq 3$.

We claim that $t_a \geq 4$ forces $t_a + p_a \leq 5$. If $t_a = 5$ and $p_a \geq 1$, choose an *aab* corner on a pair-type cube and a minority cube supporting $\{a, b, c\}$. The pair-type cube supplies one *aab*

corner and five T cubes supply $2aaa + aab + 2aac$, completing the six non-trichromatic corners of $R(a; b, c)$. If $t_a = 4$ and $p_a \geq 2$, two pair-type cubes have a common third colour b, since each supports two of the three possibilities. These two cubes supply $2aab$, and four T cubes supply $2aaa + 2aac$, where a minority cube again supports $\{a, b, c\}$. In both cases the construction above gives a solution, proving the claim.

If $t_a \leq 3$, the bucket bound gives $2t_a + p_a \leq t_a + 7 \leq 10$. If $4 \leq t_a \leq 5$, the claim gives $2t_a + p_a \leq t_a + 5 \leq 10$. Thus in all cases $2t_a + p_a \leq 10$, and summing gives

$$2A = 2n_T + 2n_P = \sum_a (2t_a + p_a) \leq 40,$$

contradicting $A \geq 22$ from (1). Therefore $M = 0$, and (2) forces $n_P = 0$. All 28 cubes are majority-class T cubes, and the four bucket bounds force each $t_a = 7$.

Finally, this conclusion holds for the original collection, not just its reduction. Every non-retained variety in Table 1 is replaced by a checkerboard; since the reduced collection contains none, no such replacement could have occurred. □

Theorem 2. *Every collection of 29 four-colour cubes contains a three-colour corner solution.*

Proof. Otherwise every 28-cube subcollection would have the form in Theorem 1. Take one such subcollection S and let v be the remaining cube. Remove from S a cube of a variety different from v, and insert v. The multiplicity of v's variety is now 1 or 8, whereas every variety in a collection described by Theorem 1 has multiplicity 0 or 7, a contradiction. □

4 Sharpness

Lemma 2. *In a cube using exactly three colours, the two orientations of a trichromatic corner occur equally often. If no trichromatic corner occurs, the cube is $Q(a; b, c)$ for some distinct a, b, c.*

Proof. For any cube coloured from a palette $\{a, b, c, d\}$, let $n(xyz)$ count corners of cyclic type xyz. Read the three directed transitions around each corner in the chosen cyclic order. Each edge between an a face and a b face contributes ab at one endpoint and ba at the other, so the total counts of these transitions agree. Corners with repeated colours contribute equally to both counts. The remaining contributions give

$$n(abc) - n(acb) + n(abd) - n(adb) = 0. \tag{3}$$

If only a, b, c occur, the last two terms vanish, proving the balance.

For the second assertion, if a colour c occurs on only one face, its four neighbours must all have the same colour a: a change from a to b around this face would produce a trichromatic corner. The opposite face is then the sole b face. Otherwise each colour occurs twice. If the two c faces are opposite, the four neighbours of either contain both a and b, forcing a change. If they are adjacent, the other three neighbours form a path containing both a and b, again forcing a change. Both cases produce a trichromatic corner. □

Remark. For the palette $\{0, 1, 2, 3\}$, write $\delta(xyz) = n(xyz) - n(xzy)$. Applying (3) to the different colour pairs gives

$$\delta(012) = \delta(031) = \delta(023) = \delta(132).$$

This common integer is the cube's *degree*, relative to the orientation Ω. Reflection reverses all cyclic orders and hence the degree. Each transposition of colour labels also reverses its sign; thus even colour permutations preserve the degree and odd ones reverse it. Table 2 shows that the two ordinary classes have degrees $+1$ and -1, while checkerboards have degree zero. Any cube missing a colour also has degree zero, recovering the balance in Lemma 2.

Theorem 3. *The threshold 29 in Theorem 2 is best possible. Exactly two labelled 28-cube collections fail, and they are mirror images.*

Proof. Take seven copies of each of the following four face tuples:

$$010203, \qquad 011213, \qquad 021223, \qquad 031323.$$

These are T cubes with repeated colours $0, 1, 2, 3$, respectively, all in the same ordinary orientation class. A target with any trichromatic corner requires both orientations by Lemma 2, whereas the collection supplies only one. A target without one is $Q(a; b, c)$; every one of its eight corners has colour a twice, so only the seven T cubes of repeated colour a are usable. No three-colour target is possible. Reflection gives the analogous example in the other orientation class. Theorem 1 excludes all other collections. □

5 Formal verification and computation

The threshold and the full classification have been formalised in Lean 4.35.0 [2] with mathlib v4.35.0 [3]. The formal model assigns arbitrary natural-number multiplicities to the 68 four-colour varieties. A physical solution selects eight distinct copies and specifies a proper rotation matching the three exposed faces at each corner of a target. Separate kernel checks establish coverage of all four-colour source and three-colour target face assignments and uniqueness of source varieties up to rotation. A general lemma, valid for any palette and allowing repeated colours, identifies cyclic corner matching with physical face matching by a proper rotation. Finite checks connect the stored corner codes to this lemma.

The formalisation proves the same threshold and classification using a related counting argument and finite Hall certificates for sufficient constructions. It treats checkerboards and the smaller ordinary class separately, whereas the written proof combines them as minority cubes. The lower bound checks a Hall deficiency for each of the 120 targets and each extremal collection; reflection exchanges the two collections. The formal proof does not enumerate all collections. Its axiom audit reports only Lean's standard `propext`, `Classical.choice`, and `Quot.sound`; it uses neither proof holes nor `native_decide`. The accompanying repository[1] contains the source, dependency pin and reproduction commands. It reuses the proper-rotation and outward-face geometry from the earlier MacMahon formalisation. No classification result from that project is used.

The proof does not depend on optimisation. A Hall-obstruction model first found the 28-cube example. After the bucket reduction, the compiled multicore CP-SAT backend also reported the maximum obstruction size as 28. That solver status is a computational check, not a separately verified optimality certificate.

Two further feasibility runs reported infeasibility for a 29-cube obstruction and for a 28-cube obstruction other than the two specified collections. Both use the reduced source family and the bucket bounds.

The accompanying script `check_three_colour_threshold.py` independently generates the 24 rotations from quarter-turns, verifies the 38-type reduction and all corner incidences, and checks the lower example against all 120 three-colour targets on the four-colour palette by bipartite matching and explicit Hall deficiencies. It also checks all 68 one-cube extensions of the lower example and 1,000 sampled collections of size 29. These samples do not establish the upper bound; the written proof does.

An expanded, bounded literature search on November 1, 2026 found no earlier answer to this threshold question, but does not establish priority. A separate AI review found no mathematical errors in the argument or physical model. These automated checks are not independent human peer review.

[1] https://github.com/jasondavies/macmahon-four-to-three

Declaration of AI-generated research and review

GPT-6-Astra (OpenAI), used through Codex, developed the research argument, computational search, Lean formalisation and manuscript under the author's direction. A separate AI reviewer checked the argument and physical model. The human author initiated and directed the research and accepts responsibility for the content of this paper and the accuracy of this disclosure. The paper has not received independent human peer review.

References

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[2] L. de Moura and S. Ullrich, *The Lean 4 Theorem Prover and Programming Language*, in Automated Deduction—CADE 28, LNCS **12699**, Springer (2021), 625–635. doi:10.1007/978-3-030-79876-5_37.

[3] The mathlib Community, *The Lean Mathematical Library*, in Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020), ACM (2020), 367–381. doi:10.1145/3372885.3373824. arXiv:1910.09336.